

2018-Ph-H1-Q11

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

An object is thrown vertically upwards with a velocity of 15 m/s.

What is the time taken to reach the highest point?

Solution:

At the highest point, vertical velocity = 0. Use:

$$v = u + at \Rightarrow 0 = 15 + (-9.8)t$$

$$t = \frac{-15}{-9.8} \approx 1.53 \text{ s}$$

Answer: 1.53 s

Revision Tip:

Throwing “up” means acceleration is negative (-9.8 m/s^2). Time to max height = time to stop. Use $v = u + at$.